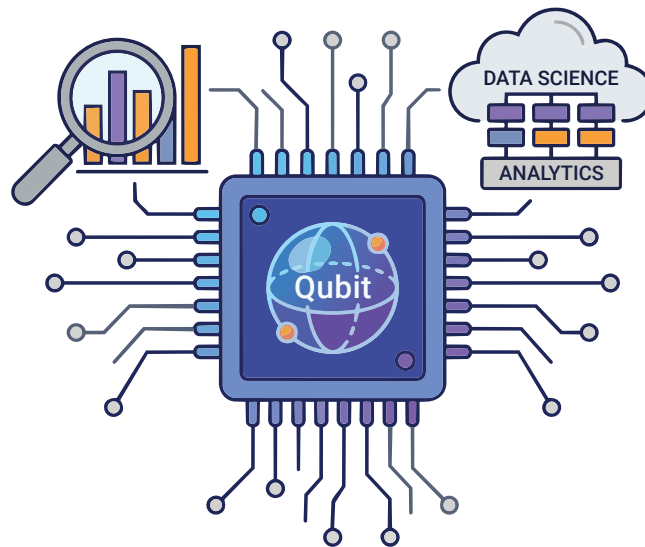


Quantum Computing in Data Science and Analytics: An Overview

Quantum computing is highly effective in processing the large amounts of data being generated by AI-based systems.



Quantum computing focuses on quantum-based information processing including cryptography, pattern recognition, strategic decision making under constraints, and many similar sensitive and performance-based applications. Table 1 lists its key uses and the algorithms developed for them.

Table 1: Key uses of quantum computing

Key task	Representative approach/Algorithm
Optimization	Quantum Approximate Optimization Algorithm (QAOA), Quantum Annealing
Cryptography	Shor's Algorithm
Simulation	Variational Quantum Eigensolver (VQE)
Search	Grover's Algorithm
Machine learning	Quantum SVM, Variational Quantum Circuit (VQC), Quantum Neural Network (QNN)

Quantum computing can be implemented in a range of domains in which supercomputing-like performance is required.

- **AI and ML:** Pattern recognition, Quantum machine learning (QML), Machine learning, Recommendation systems, Natural language processing (NLP)
- **Autonomous systems:** Autonomous vehicle decision

making, Optimization

- **Aviation:** Airline scheduling, Optimization



Figure 1: Key use cases of quantum computing in various domains